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Improvement and application of wall function boundary condition for high-speed compressible flows

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In order to develop a wall function boundary condition for high-speed flows so as to reduce the grid-dependence of the simulation for the skin friction and heat flux, a research was performed to improve the compressible wall function boundary condition proposed by Nichols. Values of parameters in the velocity law-of-the-wall were revised according to numerical experiments and the expression of temperature law-of-the-wall was modified based on theoretical analysis and numerical simulation. Besides, the formula of the heat conduction term in near-wall region was derived so that the coupling between the wall function boundary condition and CFD code was realized more accurately. Whereafter, the application study of the modified wall function was carried out. The numerical case of supersonic turbulent boundary layer on a flat plate illustrated that the modified wall function produces reasonable results of skin friction and heat flux, and profiles of velocity, temperature and turbulent eddy viscosity for coarse grids with the initial wall spacing of $y^+<400$, and that the modifications to the original wall function can obviously improve the simulation precision. As for the application of separation flows, it was found from the numerical cases of supersonic cavity flow and hypersonic axisymmetric compression corner that the compressible velocity law-of-the-wall originally established based on the fully-developed attached turbulent boundary layer approximately holds in the near-wall region inside the separation flows, which ensures that reliable skin friction and heat flux can be given by the wall function inside the separation flows, while for the region near separation and reattachment points, the wall function gives results with a relatively large error, because the velocity law-of-the-wall used in the wall function takes on obvious deviation from the real velocity profiles near the separation and reattachment points.

compressible flows, wall function boundary condition, skin friction, heat flux, numerical simulation

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1 Introduction

The accurate simulation of the skin fraction and heat flux for high-speed vehicles is a key technique of computational fluid dynamics (CFD). The velocity and temperature take on severe variations in the near-wall region of the turbulent boundary layer, which leads to tremendous difficulty for the precise prediction of the skin fraction and heat flux. Researches indicate that the computational results of skin friction and heat flux are highly sensitive to the distribution of grid points normal to the wall, especially the distance of the first grid point off the wall. It is pointed out by many researchers that [1–3] when numerically simulating the whole turbulent boundary layer, it is necessary for the distance of the first grid point off the wall to satisfy the relation $y^+=\rho_w u_\tau \Delta y/\mu_w<1$ (Δy is the distance of the first grid point off the wall, and ρ_w, u_τ, μ_w are the density, friction velocity and molecular viscosity at the wall) to ensure correct skin fraction and heat flux for high-speed vehicles. However, to satisfy $y^+<1$, Δy needs to be very small, which is generally 10^{-6} m or even smaller. Then it will lead to extremely dense grid

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points near the wall and hence huge computational consumption. For a 3-D complicated configuration, because a dense grid would greatly limit the integration timestep, the computational consumption might be too enormous to be acceptable. In addition, for the simulation of complicated flowfields, ρ_w, u_τ and μ_w are of different values in different flowfield regions, that is, Δy satisfying $y^+<1$ will be also different. Meanwhile, it is difficult to generate the mesh satisfying $y^+<1$ when ρ_w, u_τ and μ_w are unknown before computation. The above analysis illustrates that the way to obtain skin friction and heat flux by simulating the complete turbulent boundary layer is hard to be adopted in engineering problems. Consequently, the wall function boundary condition is often used in engineering applications [4], which does not resolve the near-wall region of the turbulent boundary layer while straightly uses the velocity and temperature law-of-the-wall to represent their distributions in the near-wall region and obtain the skin fraction and heat flux. The use of wall function boundary condition can relax the restriction for the density of near-wall grid points and reduce the grid-dependence in the meantime.

Most of the available widely-used compressible wall functions are originally established for the low-speed turbulent boundary layer [5–7]. The compressible velocity law-of-the-wall is generally an extension of the classic incompressible and adiabatic formula by considering the variation of the density (such as that adopted in FLUENT software), while White et al. [8] showed that the effects of compressibility and heat transfer can lead to an obvious deviation of the velocity law-of-the-wall from its incompressible form. Moreover, the uncoupled temperature law-of-the-wall is commonly used in the available compressible wall functions while the coupling effect of velocity and temperature is not included. Therefore, errors will be brought in inevitably if the low-speed incompressible wall function is directly applied to high-speed flows through a simple extension. In 2004, Nichols et al. [3] presented a new compressible wall function boundary condition including the effects of compressibility and heat transfer, which makes it possible to use wall function boundary condition to obtain skin friction and heat flux in high-speed compressible flows. Afterwards, this wall function was utilized in the numerical simulation of high-speed flows by some researchers [3,9,10].

However, we think that there are several theoretical issues in the establishment of Nichols compressible wall function, which can influence the simulation precision. Furthermore, the applicability of Nichols wall function in complex high-speed flows, such as separated flow and shock wave/boundary layer interaction, still needs to be verified in detail. In the present paper, a research to improve Nichols compressible wall function is performed to revise the theoretical issues and enhance the simulation precision. Moreover, using the modified wall function, numerical experiments for complex high-speed flows including separation and shock wave/boundary layer interaction are implemented to evaluate the applicability of the modified wall function.

2 Improvement of compressible wall function boundary condition

2.1 Theory of Nichols wall function

The compressible wall function boundary condition developed by Nichols et al. [3] is based on the following six fundamental hypotheses:

(1) Analytical expressions are available for velocity and temperature law-of-the-wall in the lower part of the turbulent boundary layer.

(2) Analytical expressions are available for the turbulent transport variables at the first grid point off the wall.

(3) There is no pressure gradient normal to the wall in the lower part of the turbulent boundary layer.

(4) The shear stress, $\tau_w=\tau=(\mu+\mu_t)(\partial u/\partial y)$, preserves constant in the lower part of the turbulent boundary layer.

(5) The heat transfer, $q_w=q=(k+k_t)(\partial T/\partial y)$, keeps constant in the lower part of the turbulent boundary layer.

(6) There are no chemical reactions, i.e., the chemistry is frozen, in the lower part of the turbulent boundary.

In traditional incompressible velocity law-of-the-wall, there are available expressions respectively for the log layer and sublayer, while no certain formula for the buffer layer. Therefore, when using the wall function, regions of the turbulent boundary layer the first grid point locates in must be verified and it is necessary to avoid the first grid point falling into the buffer layer, which will bring inconvenience to practical applications. To solve the above problem, Nichols introduced a unified form for incompressible adiabatic velocity law-of-the-wall suggested by Spalding [11]

$$y^+ = u^+ + \mathrm{e}^{-\kappa B}\mathrm{e}^{\kappa u^+} - \mathrm{e}^{-\kappa B}\left(1+\kappa u^+ + \frac{(\kappa u^+)^2}{2} + \frac{(\kappa u^+)^3}{6}\right), \quad (1)$$

where κ is Kármán constant, taken as 0.4; constant B is 5.5; u^+ and y^+ are defined as

$$u^+ = \frac{u}{u_\tau}, \quad y^+ = \frac{\rho_w u_\tau y}{\mu_w}, \quad u_\tau = \sqrt{\frac{\tau_w}{\rho_w}}. \quad (2)$$

The third term on the right-side of eq. (1) is the correction part added by Spalding. In the log layer, this term tends to $\mathrm{e}^{-\kappa B}\mathrm{e}^{\kappa u+}$ and eq. (1) reduces to $y^+=u^+$, while in the log layer, it tends to u^+ and eq. (1) reduces to $y^+=\mathrm{e}^{-\kappa B}\mathrm{e}^{\kappa u+}$, which is a transformation form of the classical incompressible logarithmic law $u^+=(1/\kappa)\ln y^+ + B$.

Based on eq. (1), Nichols introduced the compressible velocity law-of-the-wall by White et al. [8] to consider compressibility and heat transfer effects in high-speed flows:

$$u^+ = (1/2\Gamma)\left\{\beta + Q\sin\left[\phi + \frac{\sqrt{\Gamma}}{\kappa}\ln\left(\frac{y^+}{y_0^+}\right)\right]\right\}, \quad (3)$$

where

$$\Gamma=\frac{ru_\tau^2}{2c_pT_w},\qquad \beta=\frac{q_w\mu_w}{\rho_wT_wk_wu_\tau},\qquad \phi=\sin^{-1}\left(\frac{-\beta}{Q}\right),\qquad (4)$$
$$Q=(\beta^2+4\Gamma)^{1/2},\qquad y_0^+=\mathrm{e}^{-\kappa B}.$$

$r=(Pr)^{1/3}$ is the recovery factor and c_p is the specific heat at constant pressure. T_w and k_w are respectively the temperature and heat conductivity at the wall. Nondimensional parameter Γ models the compressibility effect and parameter β models the heat transfer effect. The magnitudes of the two parameters can reflect the influence on the velocity law-of the-wall by compressibility and heat transfer. Eq. (3) can naturally reduce to the corresponding form for incompressible adiabatic flows. Replacing the incompressible exponential term $\mathrm{e}^{-\kappa B}\mathrm{e}^{\kappa u+}$ in eq. (1) by eq. (3), the unified velocity law-of-the-wall including the effects of compressibility and heat transfer can be obtained as

$$y^+=u^++y^+_{\text{white}}-\mathrm{e}^{-\kappa B}\left[1+\kappa u^++\frac{(\kappa u^+)^2}{2}+\frac{(\kappa u^+)^3}{6}\right],\qquad (5)$$

where

$$y^+_{\text{white}}=\exp((\kappa/\sqrt{\Gamma})\left\{\sin^{-1}\left[\left(2\Gamma u^+-\beta\right)/Q\right]-\phi\right\})\times\mathrm{e}^{-\kappa B}.$$

In addition, the temperature law-of-the-wall is also needed to compute the heat flux. Based on the Crocco-Busemann equation [12] which describes the relationship between temperature and velocity in the boundary layer, the coupled temperature law-of-the-wall can be obtained indirectly from the velocity law-of-the-wall. For compressible turbulent boundary layer without a pressure gradient, the Crocco-Busemann equation is transformed by White [8] into

$$T=T_w\left[1+\beta u^+-\Gamma(u^+)^2\right],\qquad (6)$$

where expressions of Γ and β are given in eq. (4). For an adiabatic wall, q_w=0, that is, β=0. Then the Crocco-Busemann equation reduces to

$$T=T_w-\frac{ru^2}{2c_p}.\qquad (7)$$

Due to the coarse grids near the wall when using wall function boundary condition, turbulent transport variables should be modified at the first grid point off the wall. Turbulent eddy viscosity can be given by constant stress assumption as

$$\frac{\mu_t}{\mu_w}=1+\frac{\partial y^+_{\text{white}}}{\partial y^+}-\kappa\mathrm{e}^{-\kappa\beta}\left[1+\kappa u^++(\kappa u^+)^2/2\right]-\frac{\mu_{w+1}}{\mu_w},\qquad (8)$$

where μ_{w+1} is the molecular viscosity at the first point off the wall and the term of $\partial y^+_{\text{white}}/\partial y^+$ is given by

$$\frac{\partial y^+_{\text{white}}}{\partial y^+}=2y^+_{\text{white}}\frac{\kappa\sqrt{\Gamma}}{Q}\left[1-\frac{\left(2\Gamma u^+-\beta\right)^2}{Q^2}\right]^{-0.5}.\qquad (9)$$

It is notable that the exponent in eq. (9) is 0.5 in Nichols's original paper [3], which is found to be -0.5 based on re-derivation by the present paper.

2.2 Modifications to Nichols compressible wall function

2.2.1 Correction of the velocity law-of-the-wall

The reliability of eq. (5), velocity law-of-the-wall, is examined firstly. It is found that the uncertainty mainly comes from the values of constants κ and B. Nichols took κ=0.4 and B=5.5 according to ref. [8] by White in 1971. However, White reset κ=0.41 and B=5.0 in ref. [13] in 1991. Moreover, Wilcox also gave κ=0.41 and B=5.0 for the logarithmic law in his monograph [4]. In the present study, numerical experiments are designed to study the accuracy of eq. (5) under different values of κ and B.

The supersonic turbulent flow on a flat plate is chosen as the computational case, in which freestream parameters are typically set as M_∞=3, T_∞=288 K, ρ_∞=1.225 kg/m^3, and wall temperature is respectively taken as 300 K (cold wall), 500 K (cold wall), adiabatic wall and 900K (hot wall). The computational mesh is displayed in Figure 1, and the distance of the first point off the wall Δy is 5×10^{-7} m corresponding to $y^+\approx1$. Using such a fine mesh, the turbulent boundary layer can be completely simulated and there is no need to introduce the wall function boundary condition. The numerical simulation is performed using ACANS, a finite difference CFD code developed by us. The reliability of ACANS has been validated by a variety of simulations and the details of the code can be found in refs. [14, 15]. In the computations, Roe scheme [16] is used for discretization of inviscid flux vectors and k-ω SST model [17] is chosen to close the Reynolds stress. By analyzing White's derivation

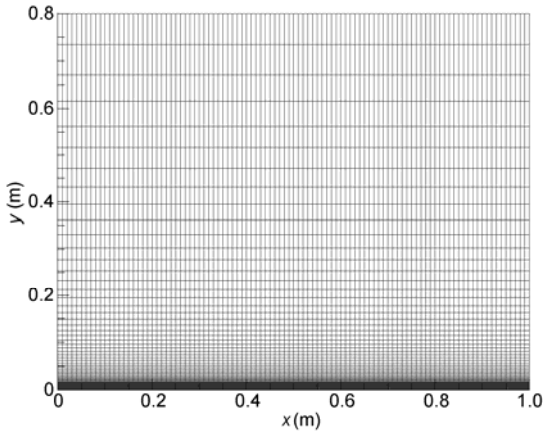


Figure 1 Computational mesh for the flat plate case.

for the compressible logarithmic law [8], it is found that the effects of compressibility and heat transfer are included through introducing the variation of mean density in the turbulent boundary layer. This physical process can be considered in numerical simulation when using the compressible form of the turbulence model. Consequently, it is reasonable to examine the reliability of eq. (5) by RANS simulation based on k-ω SST model.

Figure 2 shows comparisons of skin friction coefficient and heat flux between the computational results and Van Driest II theoretical solutions [12] under the above free-stream conditions and different wall temperatures. It can be seen that computational results agree with the theoretical solutions well, which illustrates that the simulations are reliable and hence can be used to validate the velocity law-of-the-wall. Figure 3 demonstrates the contrasts among computational velocity profiles, eq. (5) and incompressible logarithmic law at x=1 m under different wall temperature conditions. In Figure 3, different results of eq. (5) under two sets of values of κ and B are included, i.e., κ=0.41 and B=5.0 (left) as well as κ=0.4 and B=5.5 (right). Under T_w=300 K, the computational velocity profile is clearly

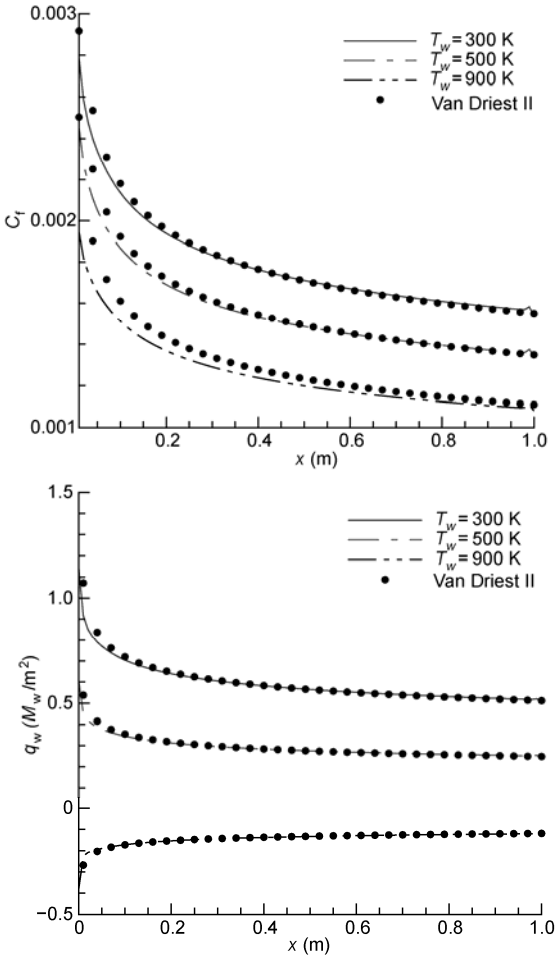


Figure 2 Comparisons between computational and theoretical results of skin friction coefficient and heat flux.

above the incompressible logarithmic law resulted from the obvious heat transfer effect from the cold wall. When T_w=500 K, the obtained velocity profile is almost consistent with the incompressible logarithmic law on account of the balance between the two effects: cold wall heat transfer which moves up the velocity profile and compressibility which moves down the velocity profile. Moreover, under the adiabatic wall condition, the velocity profile is below the incompressible logarithmic law because there is no heat transfer at the wall and compressibility effect leads to the drop. Lastly, under T_w=900 K, the velocity profile is far below the incompressible logarithmic law due to the hot wall heat transfer and compressibility effects, both of which move down the velocity profile. In addition, results from eq. (5) agree well with numerical results under cold wall conditions (T_w=300 K, 500 K) when κ=0.41 and B=5.0, while eq. (5) performs well under adiabatic and hot wall (T_w=900 K) when κ=0.4, B=5.5. Generally, there are always cold wall conditions for heat transfer calculation in high-speed flows. Therefore, κ=0.41, B=5.0 are selected in the present study, the rationality of which will be further validated in the next numerical tests.

2.2.2 *Correction of temperature law-of-the-wall*

The accuracy of eq. (6) is verified based on the above flat plate case as well. Figure 4 gives the comparisons between the computational velocity-temperature curves normal to the wall and those predicted by eq. (6) at x=1 m. All of the parameters used to calculate Γ and β in eq. (6), such as u_τ and q_w, directly come from the numerical simulations, and their accuracy has been validated in former comparisons with Van Driest II theoretical solutions.

Figure 4 indicates that velocity-temperature curves of eq. (6) fit well with numerical results near the wall, while gradually deviate from numerical results away from the wall. Large deviations emerge under cold wall conditions, and there is even an obvious mistake that T/T_e is not 1 when u/U_e=1. It is concluded that there must be some problems with eq. (6) originally from White's transformation.

Eq. (6) is deduced from the Crocco-Busemann energy integration relationship for the compressible boundary layer without pressure gradient, which is given as

$$T = T_w + \left(T_{aw} - T_w\right)\frac{u}{U_e} - \frac{ru^2}{2c_p}, \tag{10}$$

where U_e is the velocity on the boundary layer edge and T_{aw} is the adiabatic temperature at the wall, expressed as

$$T_{aw} = T_e\left(1 + \frac{\gamma - 1}{2} r Ma_e^2\right). \tag{11}$$

T_e and Ma_e are respectively the temperature and Mach number on the boundary layer edge. Figure 5 compares the temperature-velocity curves predicted by eq. (10) with the numerical results, and it is found that the two agree well. In

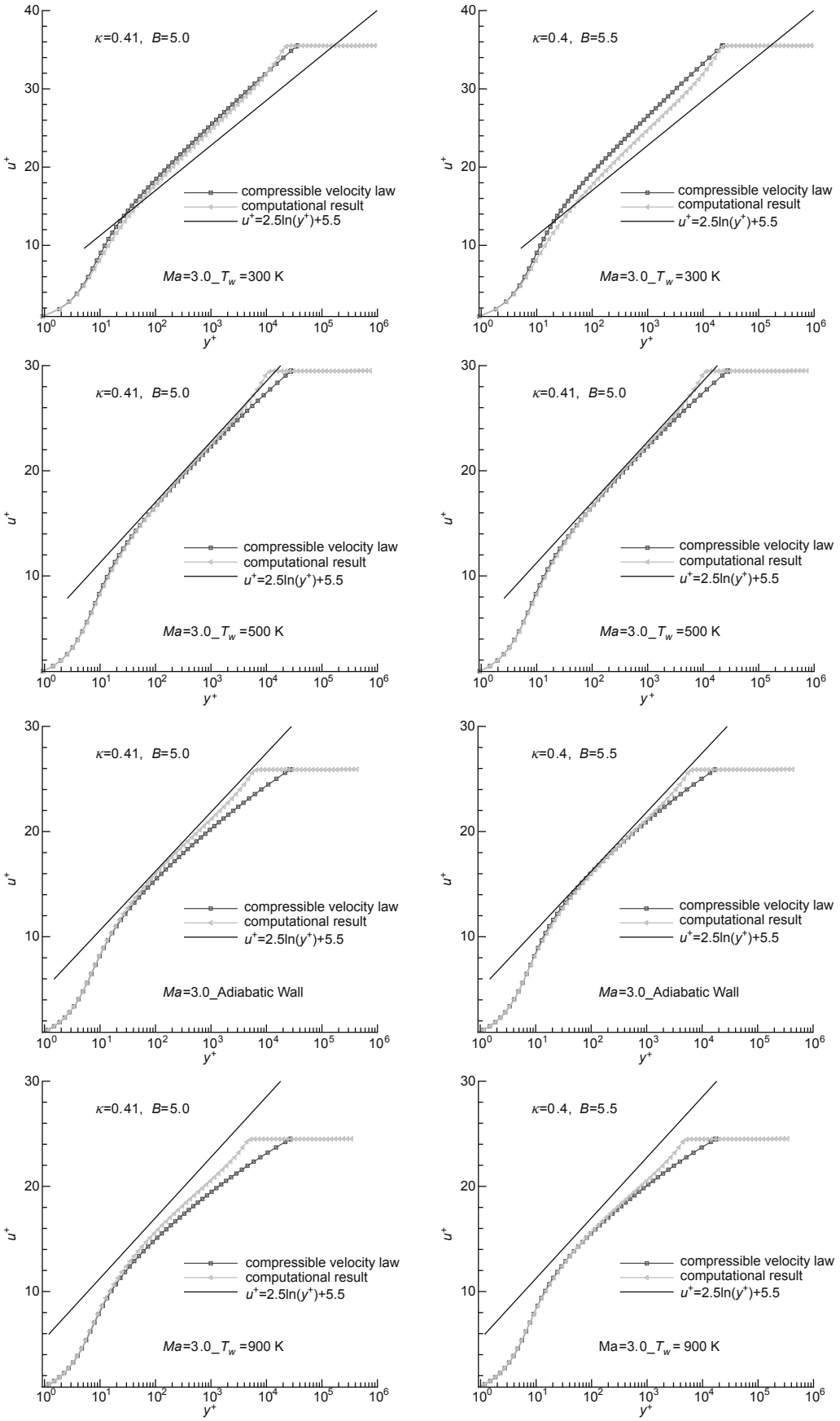


Figure 3 Comparisons between computational velocity profiles and different kinds of law-of-the wall.

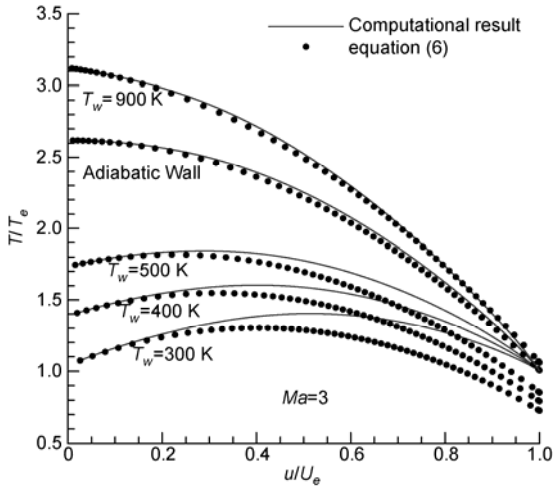


Figure 4 Velocity-temperature curves by computation and eq. (6).

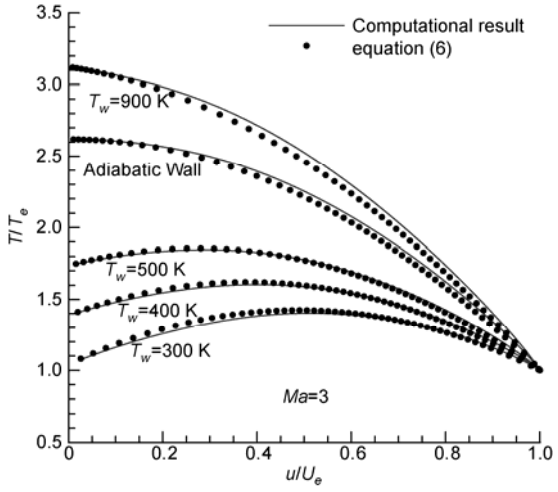


Figure 5 Velocity-temperature curves by computation and eq. (10).

White's theory [8], eq. (6) is obtained through some transformation from eq. (10), so the two should be equivalent. However, there are clear differences between the results of eq. (6) and eq. (10). Consequently, certain mistake must exist in White's transformation.

The derivation of eq. (6) is analyzed in order to revise it and make it equivalent to eq. (10). Derive the both sides of eq. (10) simultaneously, and at the wall there is

$$\frac{\partial T}{\partial y}\Big|_w = \frac{(T_{aw}-T_w)}{U_e}\frac{\partial u}{\partial y}\Big|_w, \tag{12}$$

then

$$\frac{(T_{aw}-T_w)}{U_e} = \frac{q_w\mu_w}{k_w\tau_w}. \tag{13}$$

Substitute eq. (13) into eq. (10), we have

$$T = T_w + \frac{q_w\mu_w}{k_w\tau_w}u - \frac{ru^2}{2c_p}. \tag{14}$$

By transforming u into u^+ in the above equation and noting that $\tau_w = \rho_w u_\tau^2$, eq. (6) is then obtained, where

$$\beta = \frac{q_w\mu_w}{\rho_w T_w k_w u_\tau}, \quad \Gamma = \frac{ru_\tau^2}{2c_p T_w}$$

The above derivation process shows that parameter β comes from eq. (13). After transforming eq. (13) and using the relationship $Pr=\mu c_p/k$, the expression of Stanton number is given as

$$St = \frac{q_w}{\rho_e U_e c_p (T_{aw}-T_w)} = \frac{c_f/2}{Pr}$$
$$\Rightarrow \quad \frac{St}{c_f/2} = \frac{1}{Pr} = 1.39,$$

that is, the Reynolds analogy factor is 1.39 based on eq. (13). However, the value of Reynolds analogy factor should be 1.16 determined by the large number of high-speed flow experiments and CFD results [12]. Hence, according to the more accurate value of Reynolds analogy factor, eq. (13) needs to be corrected as

$$\frac{(T_{aw}-T_w)}{U_e} = \frac{1.2q_w\mu_w}{k_w\tau_w}, \tag{15}$$

where constant 1.2 is from 1.39 divided by 1.16. Based on the above analysis, the essential reason making eqs. (6) and (10) divergent is the implicit error on Reynolds analogy factor in the expression of β. Therefore, parameter β should be expressed as

$$\beta = \frac{1.2q_w\mu_w}{\rho_w T_w k_w u_\tau}. \tag{16}$$

Then, eq. (6) can be revised by substituting eq. (16), and Figure 6 gives the comparisons of the velocity-temperature curves between the results from new version of eq. (6) and the corresponding numerical results. Contrasting Figure 5 with Figure 6, it is proved that eq. (6) has become equivalent to eq. (10) and the modification of β in the present study is correct.

In addition, it must be pointed out that the modified β expression will cause error in the region very close to the wall, because derivation in the wall normal direction on both sides of eq. (6) will lead to

$$k\frac{\partial T}{\partial y}\Big|_w = 1.2q_w. \tag{17}$$

In fact, this error is caused not by the modification of parameter β, but the error of Crocco-Busemann equation itself [18,19]. Eq. (6) by White is more accurate than eq. (10) in the region close to the wall, which can be found in comparison between Figures 4 and 6. However, eq. (6) by White leads to a large error away from the wall. Considering the

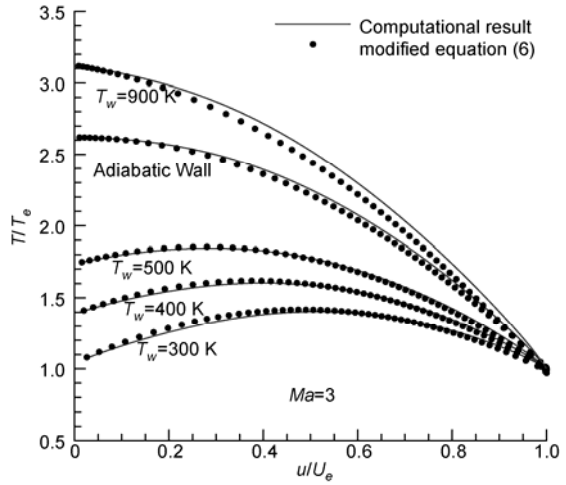


Figure 6 Velocity-temperature curves by computation and modified eq. (6).

large y^+ for the first grid point off the wall when using the wall function, it should be more accurate to select the temperature law-of-the-wall based on eq. (10), which will be further validated in the following numerical tests on the wall function boundary condition.

2.2.3 Coupling method for CFD code with wall function boundary condition

The main work for CFD code coupled with wall function boundary condition is to replace the corresponding terms of momentum and energy equations in CFD code respectively by wall shear stress τ_w and heat flux q_w obtained from the wall function.

Firstly, we use velocity law-of-the-wall (5) and the velocity u_1 at the first point off the wall obtained by CFD code to solve τ_w iteratively. Then, we replace the shear stress term $\tau_{1/2}$ of the momentum equation in CFD viscous subroutine by τ_w, as shown in Figure 7. Based on the constant stress assumption in the lower part of the turbulent boundary layer [20], it is reasonable to let $\tau_{1/2}$ be equal to τ_w.

Secondly, we utilize u^+ obtained from the above velocity law-of-the-wall and the temperature T_1 at the first point off the wall computed by CFD code to calculate parameter β by temperature law-of-the-wall (6) and further solve the wall heat flux q_w. Similarly, we replace the heat transfer term $q_{1/2}$ of the energy equation in CFD viscous subroutine by q_w, as shown in Figure 8. According to the assumption by Nichols [3] that heat transfer term $q=(k+k_t)(\partial T/\partial y)$ keeps constant in the lower part of the turbulent boundary layer, i.e. $q_{1/2}$ equals q_w. However, the typical temperature profile normal to the wall in the isothermal wall boundary layer displayed in Figure 9(a) illustrates that the temperature increase at first and then decreases along the wall normal direction, which means that there is a sign change for the $\partial T/\partial y$ value. Hence, the heat transfer term $q=(k+kt)(\partial T/\partial y)$ does not remain unchanged across the boundary layer. Furthermore, Figure 9(b)

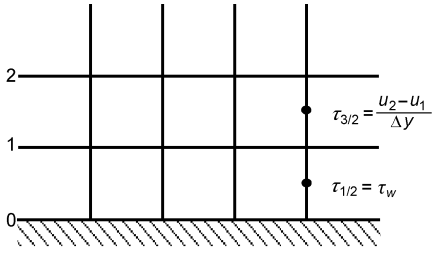


Figure 7 Shear stress near the wall in the coupling of CFD code and wall function boundary condition.

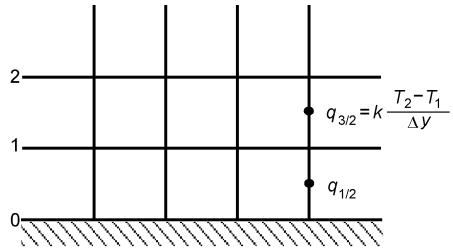


Figure 8 Heat transfer term near the wall in the coupling of CFD code and wall function boundary condition.

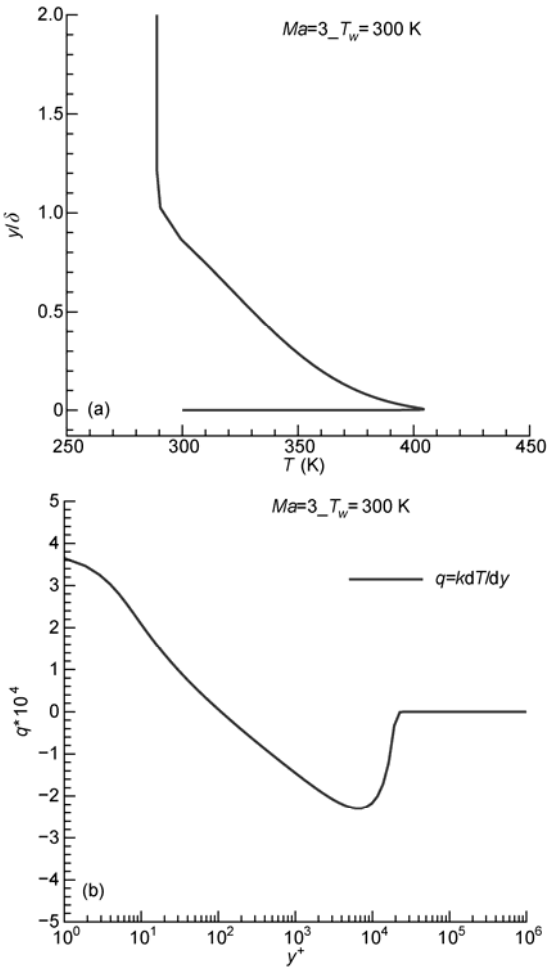


Figure 9 Typical temperature profile (a) and variation of the nondimensional heat transfer term (b).

gives the variation of $q=(k+k_t)(\partial T/\partial y)$ normal to the wall obtained by numerical simulation under fine mesh without wall function. It can be seen that this term varies continuously even very close to the wall. Therefore, the assumption by Nichols about the heat transfer term does not hold.

In the present research, based on constant stress assumption and Crocco-Busemann energy integration equation, the heat transfer term near the wall is derived as

$$q = (k + k_t)\frac{\partial T}{\partial y} = q_w - \frac{ru\tau_w}{Pr_t}, \tag{18}$$

where Pr_t is the turbulent Prandtl number, taken as 0.9. Figure 10 demonstrates the comparison of q distributions in the turbulent boundary layer respectively by numerical simulation and eq. (18). It shows that the two agree well in the lower part of the boundary layer, which proves the reliability of eq. (18). Finally, according to eq. (18), $q_{1/2}$ is given by

$$q_{1/2} = q_w - \frac{ru_{1/2}\tau_w}{Pr_t}$$

3 Code validation

A computation code is developed based on the theory of modified compressible wall function boundary condition, and incorporated into the ACANS code. Then typical test cases are selected to verify the reliability of the modified compressible wall function.

3.1 Supersonic turbulent boundary layer on a flat plate

Numerical simulations of a flat plate with isothermal wall are performed firstly using several sets of mesh with different grid scales normal to the wall in order to research the simulation accuracy of the wall function on the fully-developed attached boundary layer. In the computations, the length of flat plate is 1m, and the freestream conditions are M_∞=3, T_∞=288 K, ρ_∞=1.225 kg/m^3, T_w=300 K. As for the computational mesh, the streamwise grid points are distributed uniformly, and the space is 0.01 m. The manner of streamweise grid distribution keeps invariable when the normal grid scale varies, so the grid aspect ratio can be calculated by the corresponding streamwise and normal grid scales.

Figure 11 compares the heat flux distributions obtained by simulation using different grids with that predicted by Van Driest II theory under T_w=300 K. It can be seen that when Δy is 5×10^{-7} m, the corresponding value of y^+ is about 1, and the computational heat flux without using wall function agrees well with the Van Driest II solution. When Δy are taken as 2.5×10^{-5} m, 5×10^{-5} m, 1×10^{-4} m or 2×10^{-4} m, whose corresponding y^+ values are respectively 50, 100, 200 or 400, the results of heat flux using wall function are all close to the theoretical solution. Similarly, Figure 12 shows the computational results of the skin friction coefficients

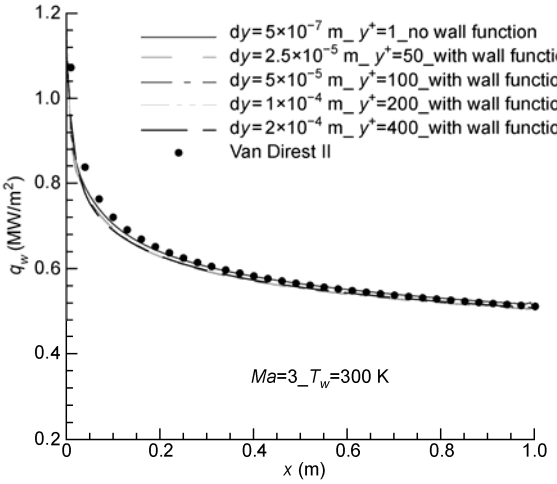


Figure 11 Heat flux distributions along the wall under different normal grid scales.

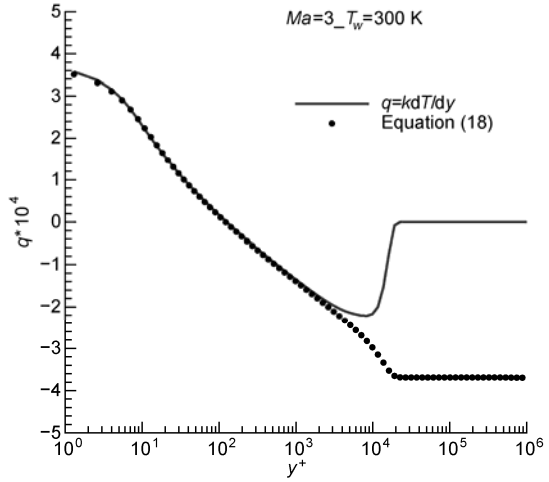


Figure 10 Nondimensional heat transfer term in boundary layer by computation and eq. (18).

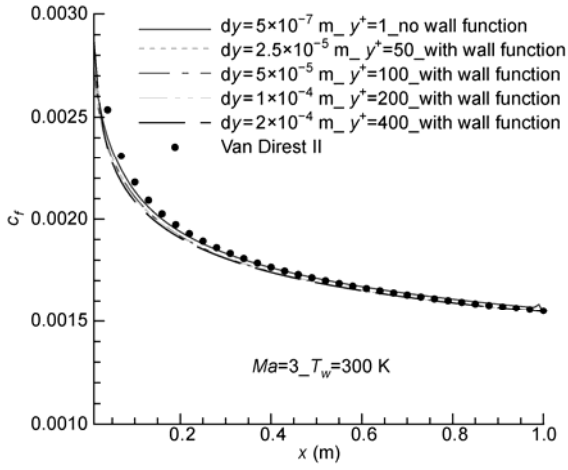


Figure 12 Skin friction coefficient distributions along the wall under different normal grid scales.

under different normal grid scales as well as the Van Driest II solutions. Likewise, the results of the skin friction coefficients under coarse grids using wall function fit well with the theoretical solution. It is indicated that the adoption of the wall function can effectively improve the grid-dependence of skin friction and heat flux, and reliable results can still be obtained even for a coarse grid with y^+ up to about 400. In addition, the variation trend of solutions by wall function with the refinement of the normal grid is given in Figure 13, from which it is illustrated that when y^+ is about 2, the solutions of wall function are almost the same as that obtained by the fine grid without wall function.

To further demonstrate the capability of resolving details of the turbulent boundary layer while using wall function in coarse grids, Figures 14, 15 and 16 respectively give the comparisons of velocity profiles, velocity-temperature curves and nondimensional eddy viscosity distributions at x=1 m obtained under different normal grid scales. It can be seen that the results of coarse grids using wall function are well consistent with the corresponding results of fine grid, which indicates that the coupling process of CFD code with the wall function boundary condition and the correction of

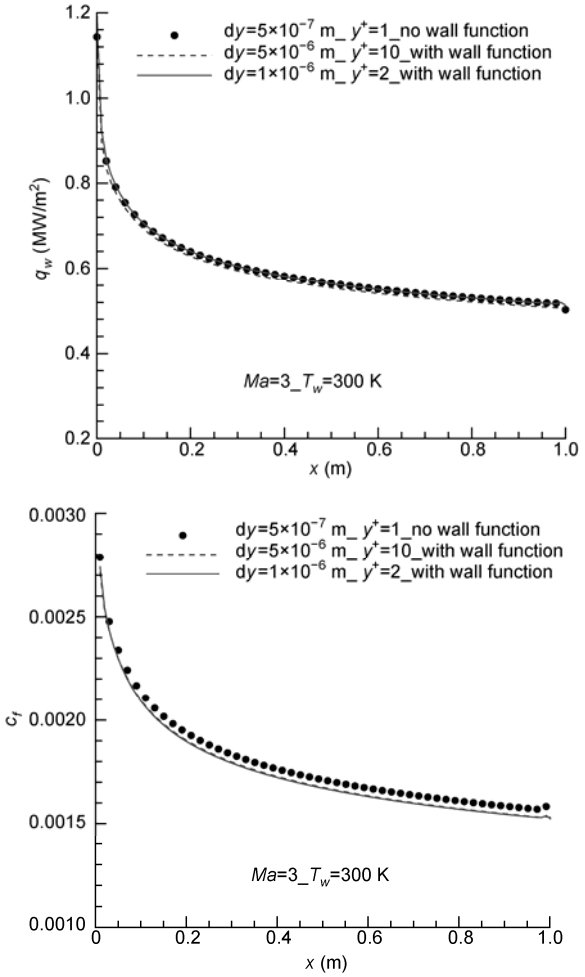


Figure 13 Variation of the solutions by wall function with the refinement of the normal grid.

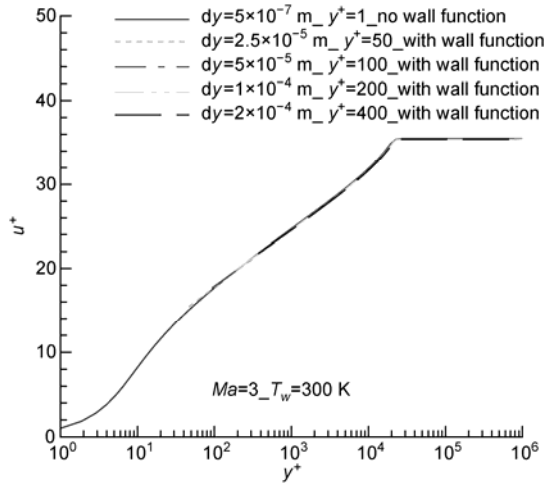


Figure 14 Velocity profiles at x=1m under different normal grid scales

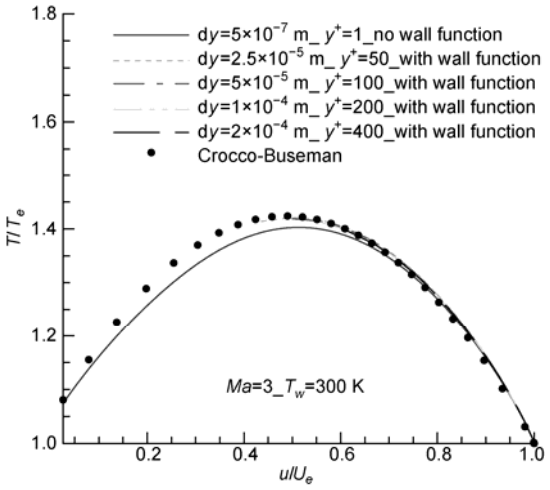


Figure 15 Velocity-temperature curves at x=1 m under different normal grid scales.

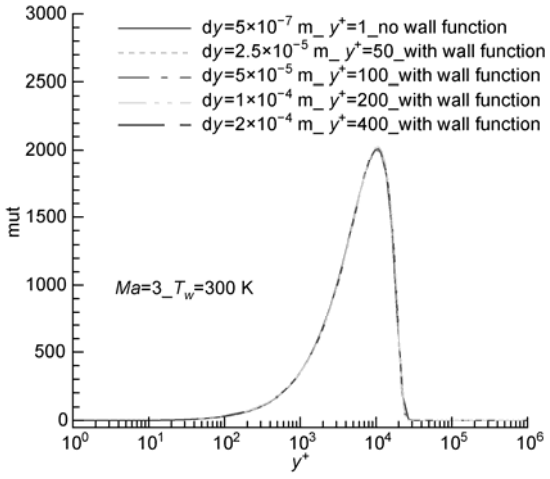


Figure 16 Nondimensional eddy viscosity variations at x=1 m under different normal grid scales.

turbulent transport variables are both reliable.

In order to examine the effect of the proposed modifications on velocity and temperature law-of-the-walls, a comparison for computation is performed on the flat plate case with T_w=300 K and Δy=1×10^{-4} m (the corresponding y^+ is 200). The results of skin friction coefficients before and after revise of velocity law-of-the-wall, and the results of heat flux before and after revise of temperature law-of-the-wall are respectively showed in Figures 17 and 18. It can be seen from Figure 17 that the skin friction coefficient is clearly lower than the theoretical solution when κ=0.4, B=5.5, while the result fits well with the theoretical solution when κ=0.41, B=5.0. Figure 18 illustrates that the heat flux distribution based on the original β expression exhibits higher than the theoretical solution, while the result using the modified β expression is consistent with the theoretical solution. Actually, Nichols gave his results of skin friction coefficient and heat flux for a flat plate in Figures 11 and 12 in ref. [3], and it can be found that his result of skin friction coefficient is lower than the theoretical solution while heat flux higher than the theoretical solution, as same as the conclusions above. Consequently, the modifications to velocity and temperature law-of-the-wall can effectively improve the results of skin fraction and heat flux obtained by wall function, which proves the rationality of the modifications.

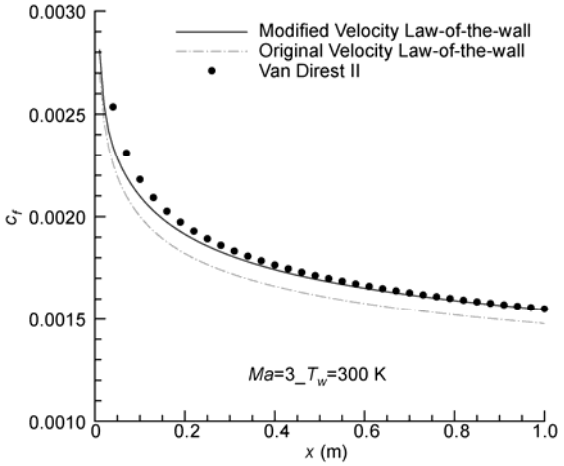


Figure 17 Comparison of skin friction coefficients before and after revise of velocity law-of-the-wall.

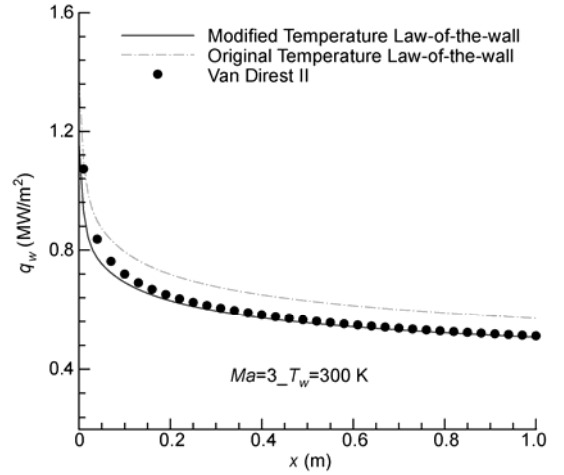


Figure 18 Comparison of heat fluxes before and after revise of temperature law-of-the-wall.

The above numerical experiments illustrate that the modified compressible wall function can effectively improve the grid-dependence of skin friction and heat flux, and produce reasonable solutions for the initial wall spacing of y^+<400 for the fully-developed attached boundary layer. Meanwhile, the wall function is able to give correct profiles of velocity, temperature and turbulent eddy viscosity under coarse grids.

3.2 Supersonic cavity flow

The wall function is established on the analytical expressions of velocity and temperature law-of-the-wall, which will fail in separated flows. Therefore, how the wall function behaves for separated flows needs to be verified, but there is no related detail research yet to the best of our knowledge. In the present study, an experimental case of supersonic cavity flow [21] is selected to conduct the evaluation on the applicability of wall function in a large separated flow. Figure 19 shows the geometric configuration of the experiment, and the freestream conditions are M_∞=2.92, p_∞=21166.9 Pa, T_∞=95.37 K, with an isothermal wall. The computational mesh is shown in Figure 20. There are several sets of mesh in the simulation, including a fine grid (Δy is 1×10^{-6} m), two coarse grids (Δy are 5×10^{-5} m and 1×10^{-4} m, respectively). The values of the aspect ratio for the first grid point off the wall in the fine grid are among 1300–2000, while for the two coarse grids, the values can be calculated according to the corresponding Δy values.

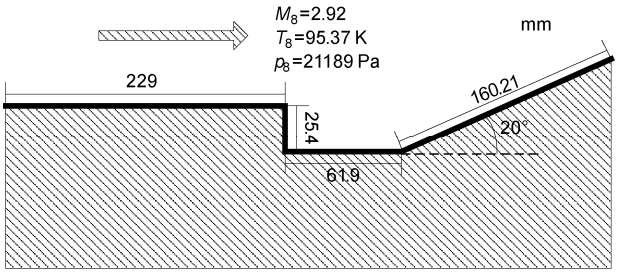


Figure 19 Geometry configuration for the supersonic cavity flow.

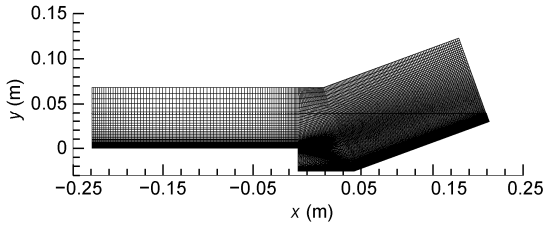


Figure 20 Computational mesh of the supersonic cavity flow.

Figure 21 gives the Mach number contour and streamlines obtained by the fine grid, in which the large separated flow and reattachment shock can be clearly seen. The flowfield characteristics of coarse grids take on no significant difference compared with those of the fine grid so the results of coarse grids are not displayed. The contrasts of pressure distributions between experimental data and the results respectively obtained by the fine grid without wall function and the coarse grids with wall function are demonstrated in Figure 22, which shows that all the computational results agree well with the experimental data. Furthermore, the slight difference between the results by fine and coarse grids is caused by a preceding reattachment point predicted by coarse grids relative to that by fine grid, which leads to a deviation of the position of reattachment shock wave.

Figure 23 shows the comparisons of skin friction coefficients between experimental data and computational results calculated by the three sets of mesh. Only after the reattachment point there is experimental data of skin friction coefficient and the result by the fine grid is consistent with the experimental data. The skin friction coefficient obtained by the coarse grid with $\Delta y=5\times10^{-5}$ m using wall function also matches with the experimental data while the result by the coarse grid with $\Delta y=1\times10^{-4}$ m takes on a little lower than the experimental data. By analyzing the skin friction coefficient results inside the separation zone in the cavity, it is found that the results obtained by coarse grids fit well with that by fine grid in most of the separation zone (x=0–0.09 m), but near the reattachment region (x=0.09–0.12 m), the results by the coarse grids with wall function show clear discrepancy with respect to the fine grid.

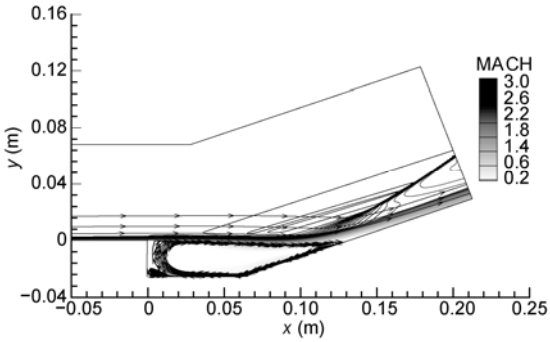


Figure 21 Mach number contour and streamlines near the cavity obtained by the fine grid.

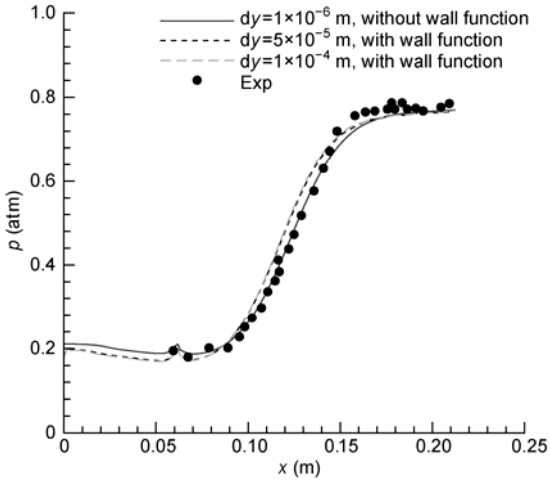


Figure 22 Pressure distributions along the wall obtained by different grids.

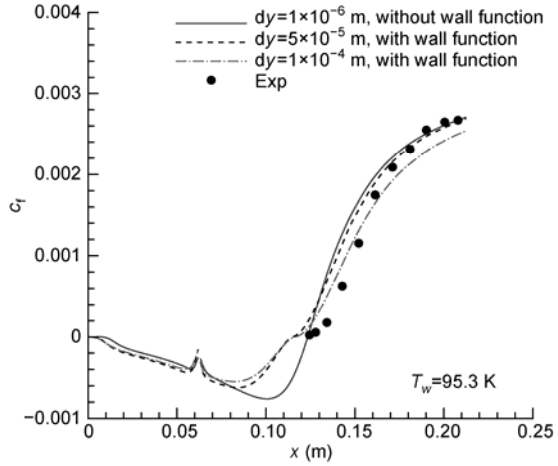


Figure 23 Skin friction coefficient distributions along the wall obtained by different grids.

Besides, the distributions of y^+ along the wall corresponding to the three grids are also given in Figure 24, and it can be seen that the variation trends of y^+ are consistent. The value of y^+ increases along the flow direction, then decreases promptly near the reattachment point, and afterwards it continuously rises. The y^+ value gives a maximum value of about 3 for the fine grid with $\Delta y=1\times10^{-6}$ m, while the maximum y^+ values for the two coarse grids with $\Delta y=5\times10^{-5}$ m and 1×10^{-4} m are up to about 200 and 400, respectively. For the coarse grids, the wall function can still give reliable distributions of skin friction coefficient.

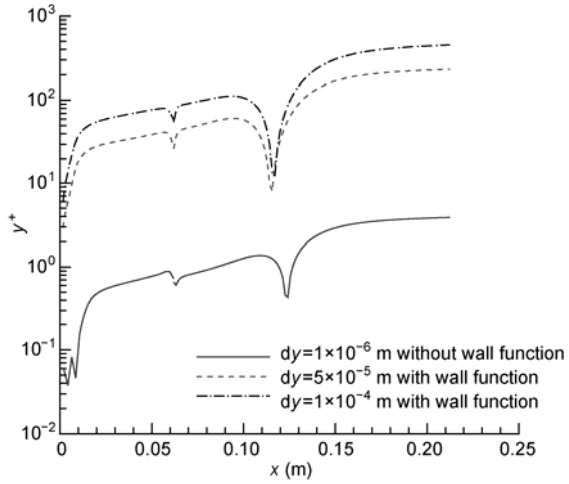


Figure 24 Distributions of y^+ values corresponding to different grids.

The derivation of wall function is based on the theory of the fully-developed attached turbulent boundary layer. Theoretically speaking, both the velocity and temperature laws-of-the-wall cannot be used in separated flow. However, Figure 23 shows that in most of the separation zone, the wall function can produce a rational result of skin friction coefficient. To perform a detailed analysis on the applicability of the wall function in separated flow, velocity profiles on several streamwise positions inside the separation zone are extracted and studied. Figure 25 compares the streamwise velocity profiles obtained by the fine grid with those predicted by the velocity law-of-the-wall (5), in which the velocity in the separation zone is taken as its absolute value. Results on 6 streamwise positions are shown in Figure 25.

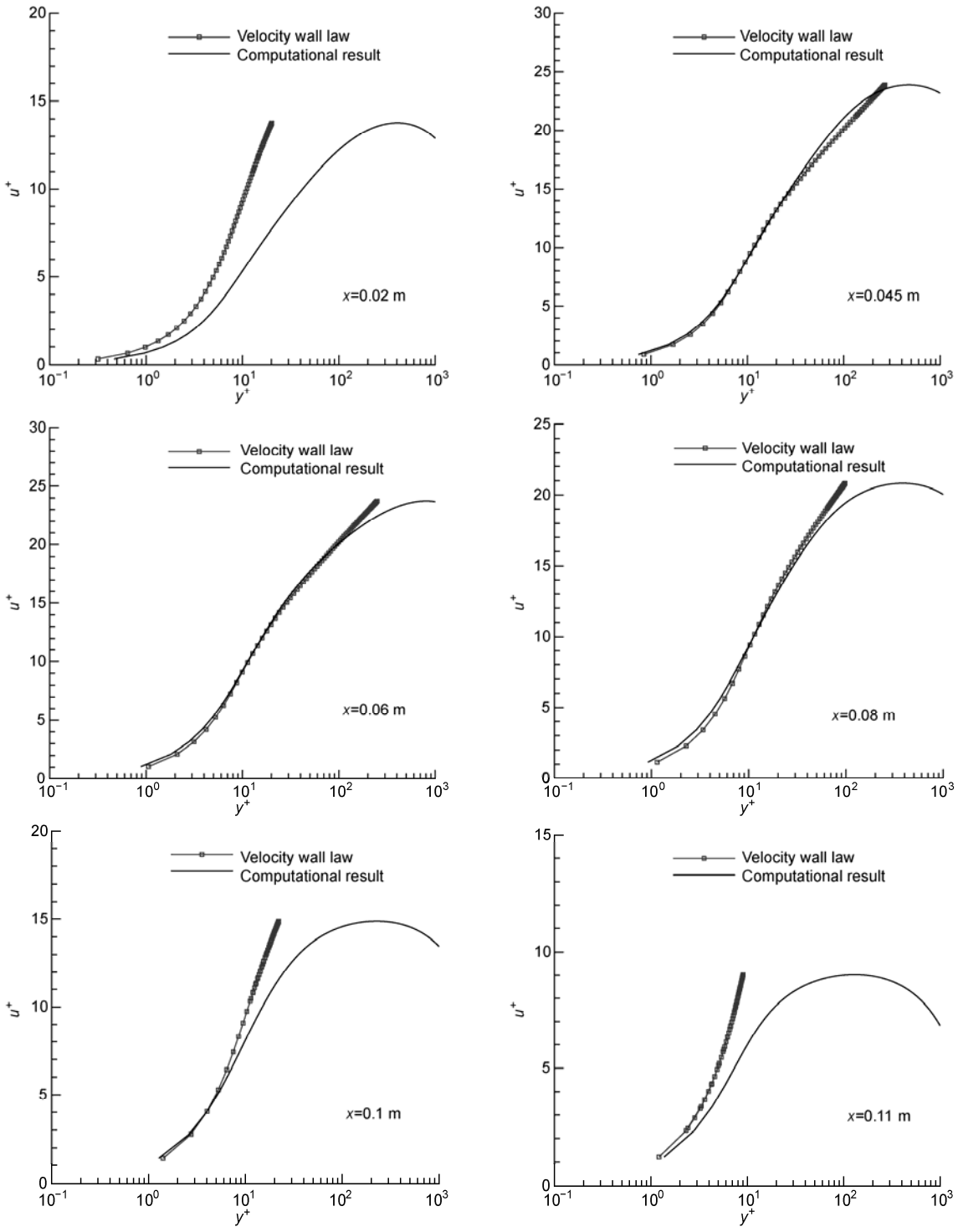


Figure 25 Comparisons between the streamwise velocity profiles obtained by the fine grid and those predicted by eq. (5).

The position of x=0.02 m is near the corner and can be considered as the separation point; the three positions of x=0.045 m, 0.06 m and 0.08 m locate inside the large separation zone, while x=0.1 m and 0.11 m are near the reattachment point. Analysis shows that inside the separation zone (x=0.045 m, 0.06 m and 0.08 m), the velocity profiles predicted by eq. (5) can maintain a good coincidence with the computational velocity profiles of separated flow (velocity is negative) near the wall (y^+ is less than 100). It is just this fact that leads to the reasonable results by wall function even in the separation zone. However, at x=0.02 m near the corner and x=0.1 m, 0.11 m near the reattachment point, the velocity profiles predicted by eq. (5) exhibit significant deviations from the computational results, even with y^+ less than 10. In this case, the deviation of eq. (5) at x=0.02 m does not lead to a large calculation error of skin friction coefficient by wall function (Figure 23). Nonetheless, the deviations of eq. (5) at x=0.1 m and 0.11 m near the reattachment point result in significant errors of the wall function, and the position of reattachment point by wall function is also inaccurate (Figure 23).

The simulation results above show that eq. (5) originally based on the theory of fully-developed attached turbulent boundary layer approximately holds for the near-wall region inside the large separation zone and can produce reasonable skin friction, while for the region near separation and reattachment points where the flow largely deviate from the fully-developed turbulent boundary layer, eq. (5) will be erroneous and the results of skin friction coefficient and reattachment point given by the wall function will present large errors.

3.3 Hypersonic flow around an axisymmetric corner

Herein, an application study of the modified wall function is implemented utilizing an experimental case of hypersonic flow around an axisymmetric corner which measures the data of wall heat flux. The experiment is from ref. [22] and the model is a hollow cylinder with a flare whose geometry configuration is shown in Figure 26. A compressible turbulent boundary layer is produced around the surface of the cylinder, and a separated flow is induced by the corner of the flare, which leads to a separation shock wave intersecting with the compression shock wave on the inclined surface of the flare. The freestream conditions are given as follows: M_∞=5.01, T_∞=83.1 K, p_∞=6561.5 Pa, T_w=300 K. The 2-D axisymmetric governing equations are solved based on Roe scheme and the k-ω SST turbulence model. In the experiment, the boundary layer transition occurs near the position of x=0.125 m. To simulate the transition in computation, the laminar simulation (μ_t is forced to be zero) is adopted before this critical position, while afterwards the turbulent simulation works.

The computational mesh is given in Figure 27. There are one fine grid (Δy is 1×10^{-6} m) and four coarse grids (Δy are

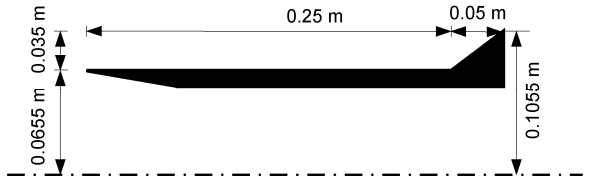


Figure 26 Geometry configuration of the hollow cylinder with flare.

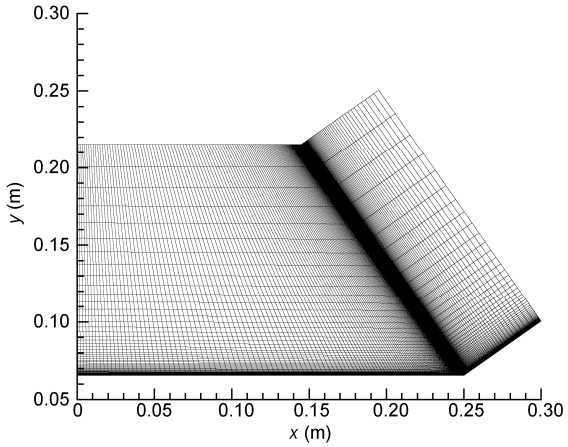


Figure 27 Computational mesh generated for the hollow cylinder with flare.

respectively 2×10^{-6} m, 1×10^{-5} m, 5×10^{-5} m and 1×10^{-4} m). Likewise, the wall function will not be employed when using the fine grid while will be adopted when using the coarse grids. The values of the aspect ratio for the first grid point off the wall in the fine grid are among 50–3500. Whereas, for the four coarse grids, the values of the aspect ratio can be calculated according to the corresponding Δy values.

Figure 28 demonstrates the Mach number contour and streamlines near the corner obtained by the fine grid. The separation shock wave, separation flow and compression shock wave are all distinctly resolved. The flowfield characteristics of coarse grids take on no significant difference compared with those of fine grid, so the results of coarse grids are not given here. The comparisons of the pressure

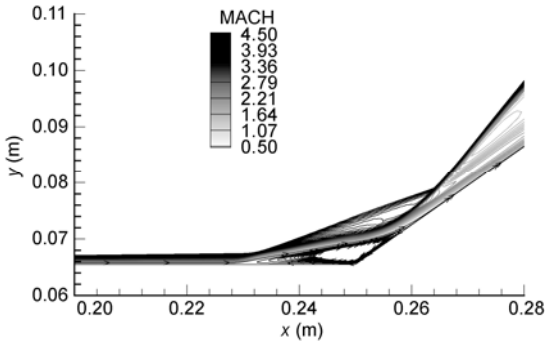


Figure 28 Mach number contour and streamlines near the corner obtained by the fine grid.

distributions on the wall obtained by different grids with experimental data are shown in Figure 29. All of the computational results totally agree well with the experimental data. By comparing the results of the fine grid and coarse grids, it is found that for the coarse grid with Δy 2×10^{-6} m, the pressure distribution is almost the same as that obtained by the fine grid, while for the other three coarse grids, deviations exist in the capture of the separation and reattachment points using wall function and the pressure distributions inside the separation region fit well with the results of fine grid.

Figure 30 shows the comparisons of the wall *St* number distributions obtained by different grids and experimental data. It can be seen that the heat flux distribution of fine grid agrees best with the experimental data. For the coarse grid with Δy 2×10^{-6} m, the obtained result using wall function is likewise consistent with that of the fine grid. As to the results of the other three coarse grids using wall function, the results are well consistent with both the results of fine mesh and experimental data before the position of the corner (x=0.25 m). Whereas in the region between the corner position and the reattachment point (x=0.25–0.27 m), the results of the coarse grids exhibit obvious deviations. For the coarse grid with Δy= 1×10^{-5} m, the computational result is acceptable, but for the other two coarse grids (Δy=5×10^{-5} m, 1×10^{-4} m), the results are obviously lower than the experimental data. However, the results of all the coarse grids agree well with the experimental data again after the reattachment point.

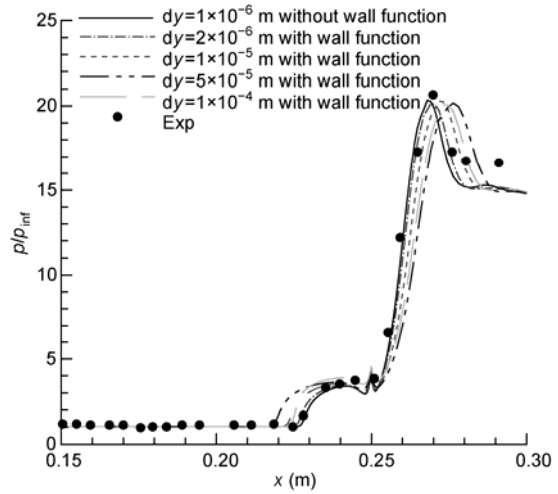


Figure 29 Pressure distributions along the wall obtained by different grids.

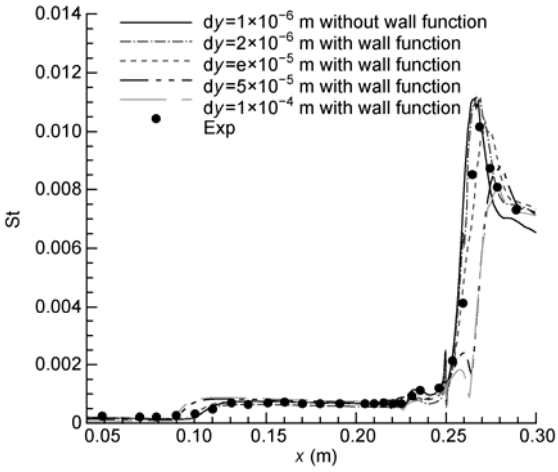


Figure 30 *St* number distributions along the wall obtained by different grids.

In summary, the above comparisons of pressure and heat flux distributions show that the results obtained by wall function under coarse grids are reliable in the attached boundary layer and internal separation region, while the heat flux results near the separation and reattachment points give obvious errors. The reason is the same as that causing the deviation of skin friction coefficient near the reattachment point in the supersonic cavity flow case of Section 3.2. Both deviations are due to the obvious difference between eq. (5) and the real velocity profile near the reattachment point. This deviation cannot be avoided unless the velocity and temperature laws-of-the-wall are established particularly for the separation and reattachment points.

4 Conclusion

A research was performed to improve the compressible wall function proposed by Nichols based on a unified law-of-the-wall and including the effects of compressibility and heat transfer. Firstly, the values of parameters in the velocity law-of-the-wall were revised according to numerical experiments. Secondly, the expression of temperature law-of-the-wall was modified based on theoretical analysis and numerical simulation. Lastly, the formula of the heat transfer term in the near-wall region was derived so that the coupling between the wall function boundary condition and CFD code is realized more accurately.

Whereafter, for the applications in typical high-speed flows such as the fully-developed attached boundary layer, separated flow and shock wave/boundary layer interaction, corresponding test cases were selected to verify the reliability of the modified compressible wall function. It was concluded that

(1) For the flat plate case, the modified compressible wall function can produce reasonable solutions of skin friction and heat flux for the initial wall spacing of y^+<400. Meanwhile, the wall function is also able to give correct profiles of velocity, temperature and turbulent eddy viscosity under coarse grids. Moreover, numerical experiments prove that the modifications to the original wall function can effectively improve the precision of skin fraction and heat flux.

(2) For the supersonic cavity flow including separation,

the compressible velocity law-of-the-wall, originally established based on the fully-developed attached turbulent boundary layer, approximately holds in the near-wall region inside the separation flows, which ensures that reliable skin friction and heat flux can be given by the wall function inside the separation flows. Meanwhile for the region near separation and reattachment points, the compressible velocity law-of-the-wall will present obvious error, leading to deviation of skin friction coefficient and reattachment point given by the wall function.

(3) For the case of hypersonic flow around an axisymmetric corner, the results of heat flux obtained by wall function under coarse grids are reliable in the attached boundary layer and internal separation region, while give obvious error near the reattachment point. The demand on the grid density near the reattachment point is relatively critical when using wall function and the normal grid points cannot be too coarse in order to obtain reasonable results of the heat flux.

To improve the simulation accuracy near the separation and reattachment points when using the wall function boundary condition, it is necessary to establish velocity and temperature laws-of-the-wall particularly for the separation and reattachment points.

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